C. elegans 11-track RNA-seq AlphaGenome Adapter Update

实验版本、结果、diagnostics 与下一步判断

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这份 slides 怎么读

我把长 checkpoint 名改成了短代号

Prefix	Meaning	Examples
A	128 bp adapter 系列	A0, A1, A2
B	1 bp-B residual correction 系列	B0, B1
S	signal-weighted loss 系列	S1, S2, S3
R	region-weighted loss 系列	R1, R2
G	gene auxiliary loss 系列	G1, G2
K	calibration 系列	K1, K2
P	downstream probe / representation 参考	P1

- 正文只用短代号，方便听汇报。
- 最后 appendix 给出短代号到完整 run directory 的对应关系。
- 所有新结果都是 validation-only；没有继续用 test 选模型。

从上次 PPT 到现在：我实际多跑了什么

Stage	Ran	What it answered
A-series	128 bp head/loss/schedule sweep	128 bp baseline 还能不能明显提高？
B-series	1 bp residual correction	1 bp embeddings 是否能在 128 bp base 上补信号？
Diagnostics	top15 extended diagnostics	当前 best 模型到底弱在哪里？
S/R/G/K	signal, region, gene-aux, calibration	能否针对 high-signal / exon / gene-level / amplitude 修正？
All-model eval	171 个已训模型全量重测	如果不只看 MSE，哪个模型更有生物 utility？
Probe	gene-profile downstream probe	输出里是否保留 gene-level / track-level 可用结构？

数据和约束没有变

- Reference: C. elegans WBcel235
- 20 sample-level RNA-seq bigWig
- grouped into 11 RNA-seq tracks
- window: 1,048,576 bp
- stride: 524,288 bp
- target: log1p RNA-seq signal

Split	Chromosome	NPZ
Train	I–IV	116
Valid	V	39
Test	X	33

Rule

后续所有训练、筛选、diagnostics 都只用 train/valid。
Test 没有继续用于调参。

A-series: 128 bp 后续 sweep

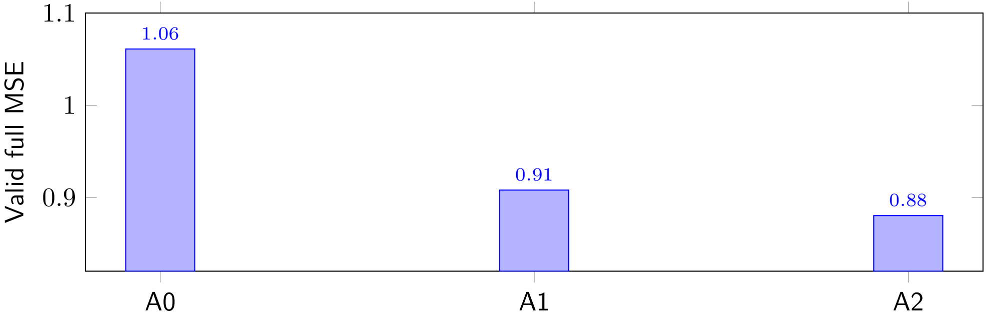
ID	Version	Valid MSE	MAE	Pearson
A0	5/18 128 bp baseline	1.0609935	0.718662	0.588099
A1	residual-conv3 sweep winner	0.9079802	0.589446	0.663840
A2	Phase4 conv5 step4000 winner	**0.8803258**	**0.584829**	**0.673917**

Analysis

128 bp **不是已经到头**。**更合适的** head、full-log1p target、hybrid/SmoothL1 loss **和** LR schedule **把** valid MSE **从** 1.06 **推到** 0.88。

A2 后来作为 1 bp residual correction 的 frozen base。

A-series: MSE 变化图



这一步回答：旧 PPT 的 128 bp baseline 不是最终水平。

B-series: 1 bp-B residual correction

ID	Version	Full MSE	MAE	Pearson	common128 MSE
A2	128 bp frozen base	0.8803258	0.584829	0.673917	0.868292
B0	1 bp-B rank1	**0.8355771**	**0.581805**	**0.694634**	0.837371
B1	1 bp-B rank3	0.8356444	0.582023	0.694629	**0.837121**

Analysis

1 bp embedding 的 residual correction 是有用的：相对 A2，B0 valid MSE 降了约 0.04475，Pearson 增加约 0.02072。

B0 是当前 full-MSE reference；B1 是 gene/pooled metrics 的 co-reference。

B-series top15：进入 plateau

top15 共同模式

- 都是 A2 frozen 128 bp prediction
- 加 1 bp residual correction
- 主要差别：warm-start、LR、beta、track weights
- top15 full-MSE spread 约 0.00137

剩余弱点

- high-signal / top1 / top5 amplitude
- exon / gene-body
- intestine T1/T3/T4
- gene-level expression
- 1 bp local-gradient / boundary

这一步回答：继续同方向小调参收益很小，要开始按弱点设计实验。

Diagnostics: B0 的具体弱点

Metric	B0 value	Meaning
high_gt3 MSE	4.9257	high expression amplitude weak
top5 true MSE	5.8610	top 5% true signal weak
top1 true MSE	10.8414	extreme peaks weak
exon MSE	1.6658	exon region error high
gene-body MSE	1.1104	gene-body signal not solved
gene Spearman	0.5835	gene rank partially preserved
local-gradient Pearson	0.3451	local boundary shape weak

Analysis

B0 的全局 MSE 很好，但 biologically important high-expression / gene-body / intestine signal 还没有完全学好。

S-series: signal-weighted loss

ID	Version	Full MSE	high_gt3	top5	top1
B0	reference	**0.8356**	4.9257	5.8610	10.8414
S1	top1x5, low LR	0.8450	4.4820	5.2960	9.8991
S2	top5x3	0.8632	**4.1605**	**4.9061**	**9.2685**
S3	intestine-highx3	0.8531	4.3510	5.1905	9.8277

Analysis

Signal weighting 确实能修 high-signal amplitude；代价是 full MSE 变差。它们不适合作为唯一主模型，但适合保留为 high-expression / intestine co-winner。

R-series: region-weighted loss

ID	Version	Full MSE	Exon	Gene-body	Intergenic	Rep score
B0	reference	**0.8356**	1.6658	1.1104	**0.2980**	0.8176
R1	exon-gene v2	0.8367	1.6591	1.1083	0.3048	**0.8881**
R2	exon-gene v1	0.8377	**1.6550**	**1.1079**	0.3084	0.8317

Analysis

Region weighting 的效果是清楚的：exon/gene-body 下降，但 intergenic 变差。R2 是最强 region winner；R1 是最平衡的 representation winner。

G/K-series: gene auxiliary 与 calibration

ID	Version	Full MSE	Gene MSE	Gene Spearman	high_gt3
B0	reference	**0.835577**	**0.712698**	0.583460	4.9257
G1	gene-aux $\lambda = 0.03$	0.835827	0.713353	0.583523	4.9179
G2	R1 + gene-aux $\lambda = 0.03$	0.837019	0.715049	0.584113	4.8248
K1	track-affine lr3e-5	0.835773	0.712705	0.583438	4.9120
K2	track-affine lr1e-4	0.836070	0.712741	0.583378	4.8931

Analysis

Gene-aux 和 calibration 都有小信号，但没有形成新的主 winner。Calibration 轻微改善 amplitude；gene-aux 没有改善 gene MSE。

Intestine T1/T3/T4 专项

ID	Version	T1/T3/T4 high_gt3	T1/T3/T4 gene Spearman	Full MSE
B0	reference	5.2666	0.5460	**0.8356**
R1	region v2	5.1702	0.5469	0.8367
S2	top5x3	4.4794	**0.5513**	0.8632
S3	intestine-highx3	**4.4591**	0.5513	0.8531

Analysis

Intestine 的 high-expression signal 更偏向 S-series。也就是说 full MSE 会惩罚它们，但它们确实学到了 B0 没表达好的 intestine 强信号。

Gene-level / pooled / boundary diagnostics

ID	Version	Gene MSE	Gene Sp.	128bp MSE	1kb MSE	Gradient P.
B0	rank1	0.7127	0.5835	0.7202	0.5451	**0.3451**
B1	rank3	**0.7126**	0.5837	**0.7201**	**0.5451**	0.3446
R1	region v2	0.7148	**0.5840**	0.7213	0.5466	0.3431

Analysis

B1 是 gene/pooled co-reference。Local-gradient 没有被任何新 objective 明显修好，说明边界/高频形状可能需要结构层面的改变。

All-model evaluation: 171 个已训模型重新测评

这一步不是重新训练主模型

- 从 runs/ 中收集 171 个已有 checkpoint。
- 对每个模型在 valid chr V 上重新跑 extended diagnostics。
- 对每个模型提取 train/valid gene-level predicted profiles。
- 训练 lightweight downstream probe：只用 train chromosomes，在 chr V 验证。

Why

我们想回答：哪个模型更会找 biologically important signal，而不是哪个模型 pixel-level MSE 最低。

All-model evaluation：结果

Role	ID	Version	Key number
MSE reference	B0	1 bp-B rank1	full MSE 0.8356
Representation winner	R1	region v2	rep score 0.8881
Region winner	R2	region v1	exon/gene-body best
High-signal winner	S2	top5x3	top1 MSE 9.2685
Intestine winner	S3	intestine-highx3	T1/T3/T4 high_gt3 4.4591
Probe co-reference	P1	broad w5	probe score 0.8581

Analysis

现在最合理的汇报方式不是单一 winner，而是 winner set：B0 做 MSE reference；R1 做主 representation candidate；S2/S3 做 high-expression/intestine co-winner。

Downstream probe: 结果怎么解释

ID	Version	top5 gene recall	T1/T3/T4 recall	track corr Sp.
B0	rank1	0.4991	0.4684	0.8771
R1	region v2	0.4999	0.4694	**0.8778**
S2	top5x3	**0.5017**	0.4713	0.8760
P1	broad w5	0.4985	0.4694	0.8768

Analysis

Probe 差距不大，所以不能单独决定模型。但它支持一个判断：S-series 对 top expressed gene retrieval 有帮助；R1 是更平衡的 overall choice。

最终我会怎么给合作者解释

- 5/18 的 A0 baseline 已经过时：128 bp 后续优化到 A2，1 bp residual 到 B0。
- B0 是当前 pixel-level full-MSE reference。
- Diagnostics 显示 B0 的弱点集中在 high-signal amplitude、exon/gene-body、intestine T1/T3/T4 和 local boundary。
- S-series 证明 high-expression amplitude 可以被修，但 full MSE 会变差。
- R-series 证明 exon/gene-body 可以被修；R1 是最平衡的 representation candidate。
- 171-model benchmark 和 probe 让我们从“刷 MSE”转成“按生物 utility 选模型”。

Next steps

Model side

- 保留 B0 作为 MSE reference
- 用 R1 做 main representation candidate
- 用 S2/S3 做 high-expression/intestine co-winner
- 下一步尝试 amplitude calibration / target parameterization

Evaluation side

- 固定 gene retrieval 作为 downstream metric
- 单独跟踪 intestine T1/T3/T4
- local-gradient 作为 boundary diagnostic
- 暂时不碰 test split

Appendix: version mapping 1

ID	Short name	Human-readable version
A0	5/18 128 bp baseline	formal 128 bp adapter, 5000 steps, lr $3e{-}4$
A1	residual-conv3 sweep	128 bp residual-conv3, full-log1p, SmoothL1, lr $1e{-}3$
A2	128 bp Phase4 conv5	128 bp conv5 h256, hybrid loss, step LR decay at step 4000
B0	1 bp-B rank1	frozen A2 base + 1 bp residual conv5, hard15, beta0.5, lr $1e{-}5$
B1	1 bp-B rank3	same family as B0, different warm-start / track-weight setting

Exact run directories are recorded in docs/rna_seq11_adaptive_objective_screen_20260529.md and docs/rna_seq11_all_representation_utility_20260530.md.

Appendix: version mapping 2

ID	Short name	Human-readable version
S1	signal top1x5	B0 warm-start, hybrid alpha0.75 beta0.5, top1 signal weighted $5\times$, lr $3e{-}6$
S2	signal top5x3	B0 warm-start, top5 signal weighted $3\times$, lr $1e{-}5$
S3	signal intestine-high	B0 warm-start, intestine high-signal weighted $3\times$, lr $1e{-}5$
R1	region exon-gene v2	B0 warm-start, exon/gene-body/promoter weighted gently, 1000 steps
R2	region exon-gene v1	B0 warm-start, stronger exon/gene-body weighting, 1500 steps

Appendix: version mapping 3

ID	Short name	Human-readable version
G1	gene-aux rank1	B0 warm-start + gene-body mean auxiliary loss $\lambda = 0.03$
G2	region v2 + gene-aux	R1 warm-start + gene-body mean auxiliary loss $\lambda = 0.03$
K1	track-affine lr3e-5	B0 + identity-initialized per-track affine output calibration
K2	track-affine lr1e-4	same as K1 but higher LR and shorter run
P1	probe co-reference	1 bp-B broad w5 checkpoint; strongest near-frontier downstream-probe score